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PIZZA SALES ANALYSIS





Introduction

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This project focuses on analyzing pizza sales data using Microsoft SQL Server to extract actionable business insights from transactional records. By leveraging relational tables—including Orders, Order_Details, Pizzas, and Pizza_Types—the analysis evaluates sales performance and customer purchasing behavior. Advanced SQL techniques such as multi-table JOINS, aggregations, Common Table Expressions (CTEs), and Window Functions are applied to generate meaningful, data-driven business insights.



Objective



- Sales & Revenue Analysis: Evaluate total revenue, average order value, and daily sales trends.
- Product Performance Analysis: Identify top-selling and least-ordered pizza types, categories, and sizes.
- Operational & Temporal Insights: Analyze hourly ordering patterns and peak sales days to support operational planning.
- SQL & Reporting Skills: Demonstrate practical SQL skills through complex queries, data analysis, and business reporting.



Dataset Overview

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The Pizza Sales dataset consists of four relational tables containing information about orders, pizzas, categories, pricing, and sales quantities.

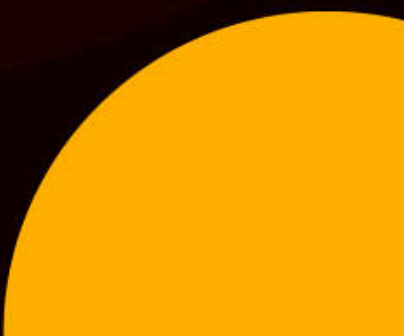
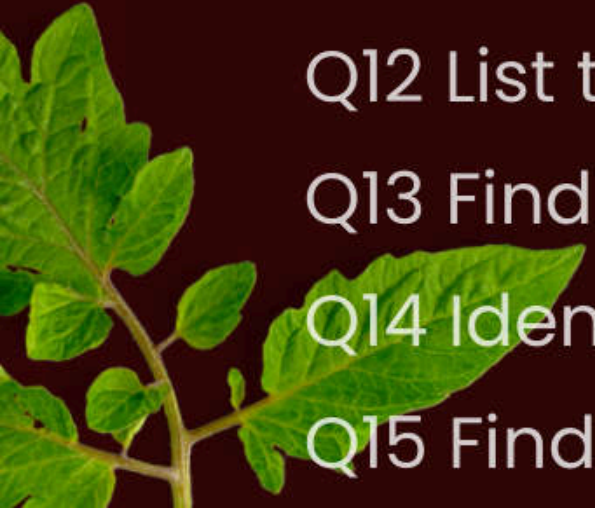
- Orders:- Order_ID, Date, Time
- Order_Details:- Order_Details_ID, Order_ID, Pizza_ID, Quantity
- Pizzas:- Pizza_ID, Pizza Type_ID, Size, Price
- Pizza Types:- Pizza_Type_ID, Name, Category, Ingredients



Business Question

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- Q1 Retrieve all orders placed on 2015-01-01.
- Q2 List all distinct pizza categories available.
- Q3 Find total number of pizzas ordered.
- Q4 Get the top 5 most frequently ordered pizza types.
- Q5 Show order_id and total pizzas ordered in each order.
- Q6 Find the number of orders placed in each hour of the day.
- Q7 Find the day of the week on which the maximum number of orders were placed.
- Q8 Find the earliest and latest order time for each day.
- Q9 Find the total revenue generated from each pizza type.
- Q10 Which pizza category generated the maximum revenue?
- Q11 Find the average number of pizzas ordered per order.
- Q12 List the top 3 orders which spent the most.
- Q13 Find the cumulative revenue generated throughout the day (running total).
- Q14 Identify the pizza that has been ordered the least number of times.
- Q15 Find the percentage contribution of each pizza category to total revenue.





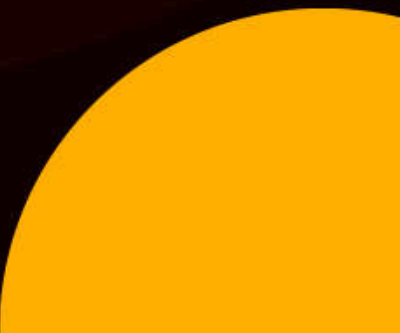
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Q1 Retrieve all orders placed on 2015-01-01.

```
Select
    order_id,
    date
From Orders
Where date = '2015-01-01'
```

Results Messages		
	order_id	date
1	1	2015-01-01
2	2	2015-01-01
3	3	2015-01-01
4	4	2015-01-01
5	5	2015-01-01
6	6	2015-01-01
7	7	2015-01-01
8	8	2015-01-01
9	9	2015-01-01
10	10	2015-01-01
11	11	2015-01-01
12	12	2015-01-01





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Q2 List all distinct pizza categories available.

```
Select  
    Distinct category  
From Pizza_Types
```

Results		Message
	category	
1	Chicken	
2	Classic	
3	Supreme	
4	Veggie	





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Q3 Find total number of pizzas ordered.

```
Select  
    SUM(quantity) as Total_Order  
From Order_Details
```

Results		Messages
	Total_Order	
1	49574	





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Q4 Get the top 5 most frequently ordered pizza types.

```
Select
    Top 5 PT.name,
    SUM(OD.quantity) as Total_Order
From Order_Details OD
join Pizzas P on OD.pizza_id = P.pizza_id Join
Pizza_Types PT on P.pizza_type_id = PT.pizza_type_id
Group by PT.name
Order by Total_Order desc
```

Results Messages		
	name	Total_Order
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Pizza	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371





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Q5 Show order_id and total pizzas ordered in each order.

```
Select
    OD.order_id,
    SUM(OD.quantity) as Total_Order
From Order_Details OD
Group by OD.order_id
Order by OD.order_id
```

Results Messages		
	order_id	Total_Order
1	1	1
2	2	5
3	3	2
4	4	1
5	5	1
6	6	2
7	7	1
8	8	1
9	9	9
10	10	2
11	11	4
12	12	1





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Q6 Find the number of orders placed in each hour of the day.

```
Select
    Datepart(HOUR, O.time) as Hour,
    COUNT(OD.quantity) as Qty
From Orders O Join
Order_Details OD on O.order_id = OD.order_id
Group by Datepart(HOUR, O.time)
Order by Hour
```

Results		Messages
	Hour	Qty
1	9	4
2	10	17
3	11	2672
4	12	6543
5	13	6203
6	14	3521
7	15	3170
8	16	4185
9	17	5143
10	18	5359
11	19	4350
12	20	3487





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Q7 Find the day of the week on which the maximum number of orders were placed.

```
Select
    Top 1 DATENAME(WEEKDAY, O.date) as Weeekday,
    COUNT(OD.quantity) as Total_Order
From Orders O Join Order_Details OD on O.order_id = OD.order_id
Group by DATENAME(WEEKDAY, O.date)
Order by Total_Order desc
```

Results		Messages
	Weeekday	Total_Order
1	Friday	8106





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Q8 Find the earliest and latest order time for each day.

```
Select
    date,
    MIN(time) as Earliest_OrderTime,
    MAX(time) as Latest_OrderTime
From Orders
Group by date
```

Results		Messages	
	date	Earliest_OrderTime	Latest_OrderTime
1	2015-02-14	11:48:35.0000000	22:52:37.0000000
2	2015-01-22	11:34:52.0000000	23:02:31.0000000
3	2015-12-20	11:55:46.0000000	22:25:20.0000000
4	2015-04-01	11:17:06.0000000	22:41:48.0000000
5	2015-03-09	11:34:01.0000000	21:58:39.0000000
6	2015-07-27	11:34:12.0000000	22:23:58.0000000
7	2015-08-02	12:01:58.0000000	22:12:31.0000000
8	2015-08-19	11:24:03.0000000	22:30:27.0000000
9	2015-08-25	11:46:31.0000000	22:17:58.0000000
10	2015-06-11	11:25:26.0000000	21:43:38.0000000
11	2015-11-27	11:21:54.0000000	22:36:01.0000000
12	2015-07-04	12:20:55.0000000	23:04:40.0000000





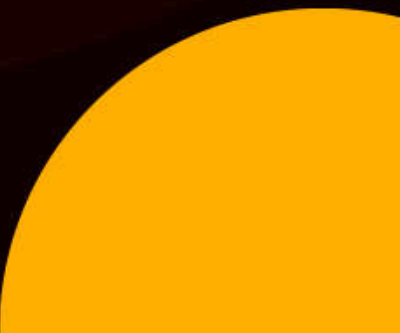
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Q9 Find the total revenue generated from each pizza type.

```
Select
    PT.name ,
    ROUND(SUM(OD.quantity * P.price),2) as Total_revenue
From Order_Details OD Join
Pizzas P on OD.pizza_id = P.pizza_id Join
Pizza_Types PT on P.pizza_type_id = PT.pizza_type_id
Group by PT.name
order by Total_revenue desc
```

Results Messages		
	name	Total_revenue
1	The Thai Chicken Pizza	43434.25
2	The Barbecue Chicken Pizza	42768
3	The California Chicken Pizza	41409.5
4	The Classic Deluxe Pizza	38180.5
5	The Spicy Italian Pizza	34831.25
6	The Southwest Chicken Pizza	34705.75
7	The Italian Supreme Pizza	33476.75
8	The Hawaiian Pizza	32273.25
9	The Four Cheese Pizza	32265.7
10	The Sicilian Pizza	30940.5
11	The Pepperoni Pizza	30161.75
12	The Greek Pizza	28154.1





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Q10 Which pizza category generated the maximum revenue?

```
Select
    Top 1 PT.category,
    Round(SUM(OD.quantity * P.price),2) as Total_revenue
From Order_Details OD Join
Pizzas P on OD.pizza_id = P.pizza_id Join
Pizza_Types PT on P.pizza_type_id = PT.pizza_type_id
Group by PT.category
order by Total_revenue desc
```

Results			Messages		
	category	Total_revenue			
1	Classic	220053.1			





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Q11 Find the average number of pizzas ordered per order.

```
With CTE as (  
  Select  
    order_id ,  
    SUM(quantity) as Total_pizza  
  From Order_Details  
  Group by order_id  
)  
Select  
  AVG(Total_pizza) as AVG_Pizzas_Per_Order  
From CTE
```

Results		Messages	
		AVG_Pizzas_Per_Order	
1	2		





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Q12 List the top 3 orders which spent the most.

```
Select  
    Top 3 OD.order_id,  
    Round(SUM(OD.quantity * P.price),3) as Total_Order  
From Order_Details OD Join  
Pizzas P on OD.pizza_id = P.pizza_id  
Group by OD.order_id  
Order by Total_Order desc
```

Results		Messages
	order_id	Total_Order
1	18845	444.2
2	10760	417.15
3	1096	285.15





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Q13 Find the cumulative revenue generated throughout the day (running total).

```
Select
    O.date,
    O.time,
    SUM(OD.quantity * P.price) as Total_Revenue,
    SUM(SUM(OD.quantity * P.price)) Over(Partition by O.date Order by O.time) as Cumm_SUM
from Order_Details OD Join
Orders O on OD.order_id = O.order_id Join
Pizzas P on P.pizza_id = OD.pizza_id
Group by O.date, O.time
Order by O.date, O.time
```

Results Messages				
	date	time	Total_Revenue	Cumm_SUM
1	2015-01-01	11:38:36.0000000	13.25	13.25
2	2015-01-01	11:57:40.0000000	92	105.25
3	2015-01-01	12:12:28.0000000	37.25	142.5
4	2015-01-01	12:16:31.0000000	16.5	159
5	2015-01-01	12:21:30.0000000	16.5	175.5
6	2015-01-01	12:29:36.0000000	24.75	200.25
7	2015-01-01	12:50:37.0000000	12.5	212.75
8	2015-01-01	12:51:37.0000000	12.5	225.25
9	2015-01-01	12:52:01.0000000	143.25	368.5
10	2015-01-01	13:00:15.0000000	41	409.5
11	2015-01-01	13:02:59.0000000	73.5	483
12	2015-01-01	13:04:41.0000000	70.75	553.75

Query executed successfully





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Q14 Identify the pizza that has been ordered the least number of times.

```
Select
    Top 1 Pt.name,
    COUNT(OD.order_id) as Total_Order
From Pizza_Types PT
join Pizzas P on
Pt.pizza_type_id = P.pizza_type_id
join Order_Details OD on
P.pizza_id = OD.pizza_id
Group by Pt.name
Order by Total_Order
```

Results Messages		
	name	Total_Order
1	The Brie Carre Pizza	480





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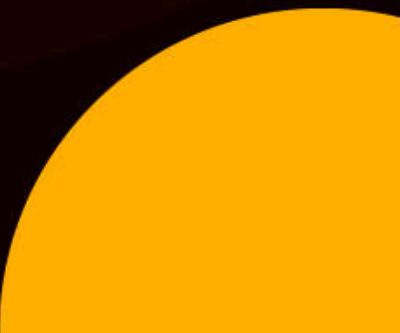
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Q15 Find the percentage contribution of each pizza category to total revenue.

```
Select
    PT.category,
    Round(SUM(OD.quantity * P.price),2) as Total_revenue,
    SUM(Round(SUM(OD.quantity * P.price),2)) Over() as Grant_Total_Sales,
    Round((SUM(OD.quantity * P.price) * 100.0) / SUM(SUM(OD.quantity * P.price)) Over(),2) as Contribution
From Pizza_Types PT join Pizzas P on
Pt.pizza_type_id = P.pizza_type_id join Order_Details OD on
P.pizza_id = OD.pizza_id
Group by Pt.category
Order by Contribution desc
```

[Results](#)[Messages](#)

	category	Total_revenue	Grant_Total_Sales	Contribution
1	Classic	220053.1	817860.05	26.91
2	Supreme	208197	817860.05	25.46
3	Chicken	195919.5	817860.05	23.96
4	Veggie	193690.45	817860.05	23.68



Insights

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- \$817.86K revenue generated from 21.35K orders.
- 49.57K pizzas sold with 2.32 pizzas per order.
- Classic Deluxe Pizza is the top-selling pizza by quantity.
- Thai Chicken Pizza generates the highest revenue.
- Classic Category contributes the largest share of total revenue.
- Large pizzas are the most preferred size.
- Peak order volumes are recorded between 12:00 PM – 1:00 PM and 5:00 PM – 6:00 PM.
- Friday & Saturday show high order demand.
- Brie Carre Pizza is the lowest-performing product.



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